

Framing AI in the Hollywood Strike:

A Social Media Analysis of Online Communities

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Academic Year: 2025/2026



May 2, 2023: WGA Calls Strike

First simultaneous WGA + SAG-AFTRA strike since 1960

WGA: Writers Guild of America. Two independent American sister labor unions representing the creators of the content (screenwriters, scriptwriters, and authors).

SAG-AFTRA : The Screen Actors Guild-American Federation of Television and Radio Artists. An American labor union representing the on-camera and voice talent (actors, broadcast journalists, and voice artists)

148 days. 160,000 workers



At the center of the debate:
Generative AI





Many possible interpretations

How did online communities frame the role of AI during the Hollywood Strike, and did framing vary systematically across communities and across the temporal phases of the dispute?

Replacement Anxiety

Workers' Rights

Creative Authenticity

Corporate Power

Technological Opportunity



A multi-method approach

399 posts

11,048 comments

5 subreddits

March–December 2023

01

Frame Detection

02

Network Analysis

03

Sentiment Analysis

04

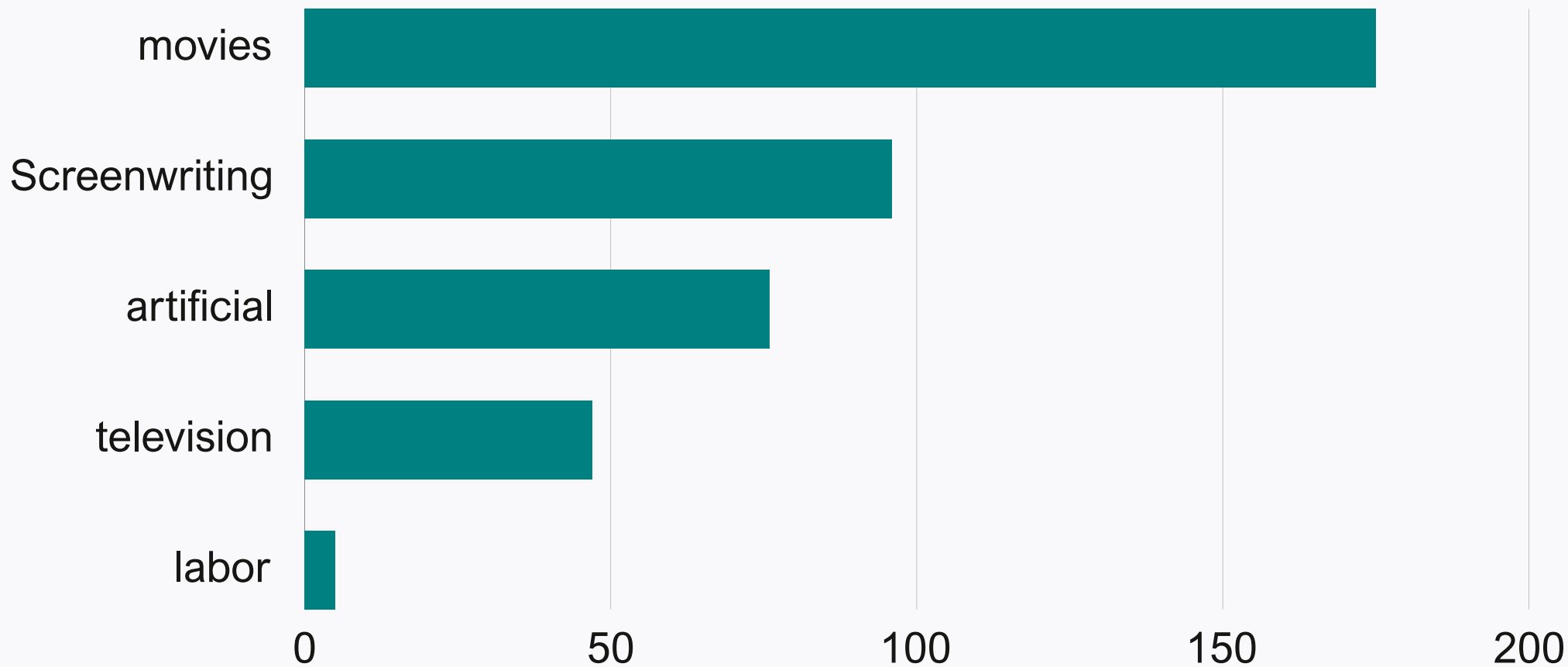
Named Entity Recognition

05

Integrate Community Analysis

Data Collection

PullPush historical Reddit archive March – December 2023, 19 search queries combining: strike-related terms + AI themes



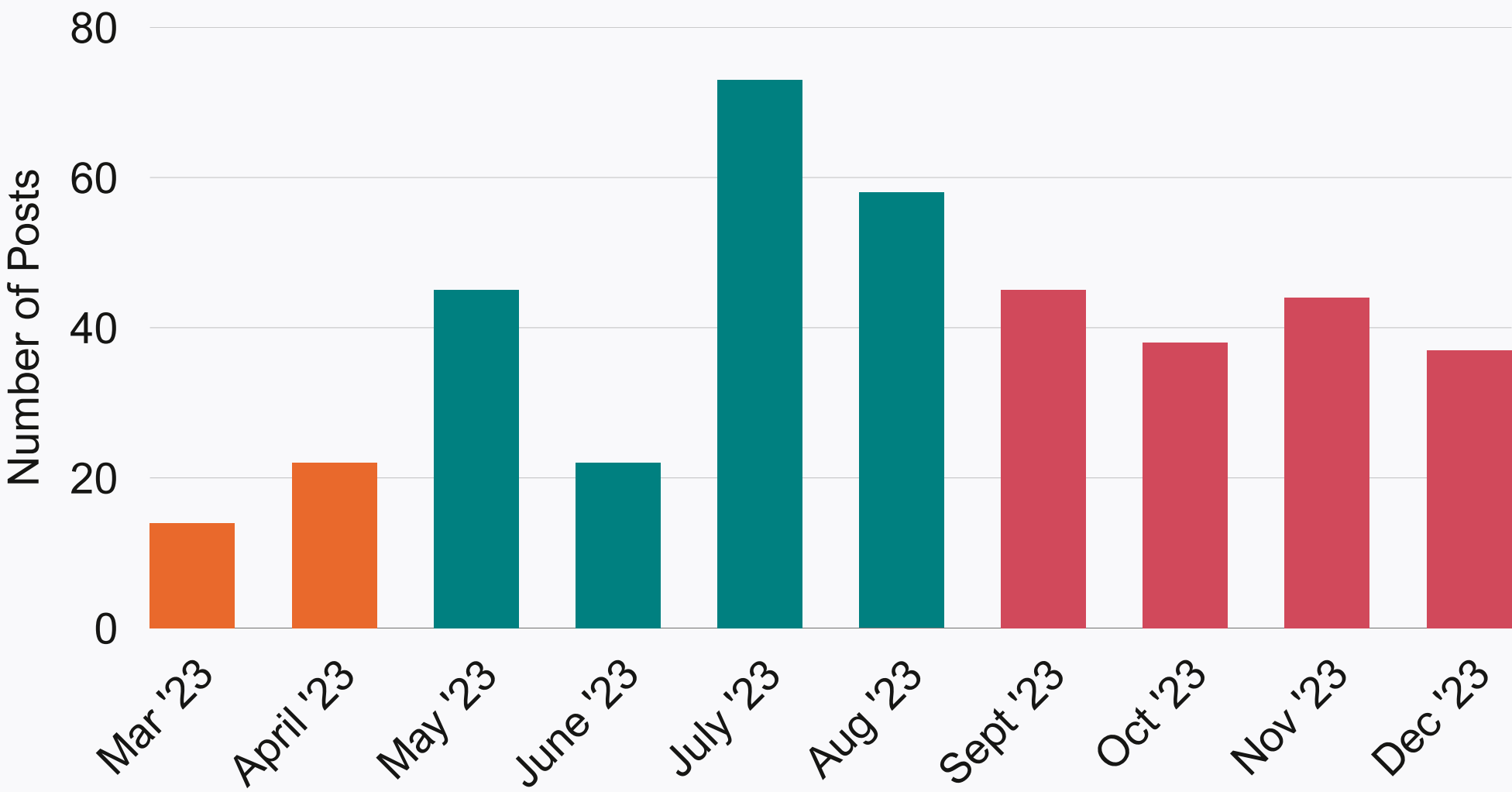
5 subreddit

01	r/movies (175)	43.9%
01	r/Screenwriting (96)	24.1%
01	r/artificial (76)	19.1%
01	r/television (47)	11.8%

Corpus Description

399 submissions
11,048 comments
239 unique authors

Pre-strike: March–April
Strike: May–August
Post-strike: September–December



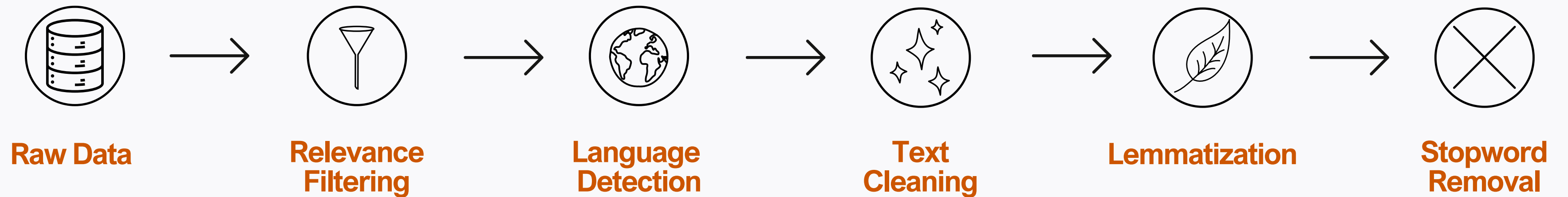
Three Temporal Phases

01	Pre-Strike (36)	9%
02	Strike (199)	49.9%
03	Post-Strike (164)	41.1%

Discussion peaked in July 2023

Data Preprocessing

A preprocessing pipeline was applied to both submissions and comments.



Standardized corpus for content and network analysis

Frame Detection



Replacement Anxiety



Authenticity & Creativity



Workers' Rights



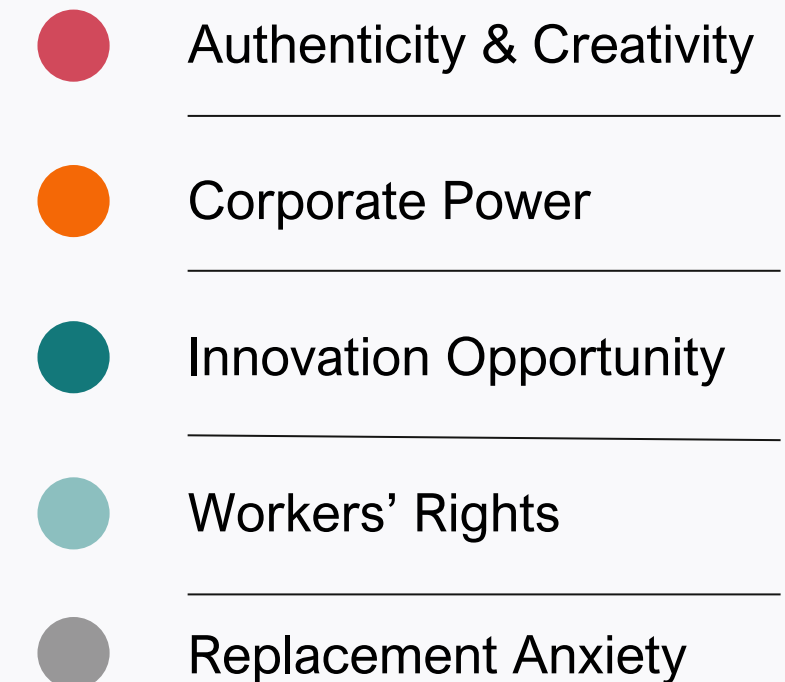
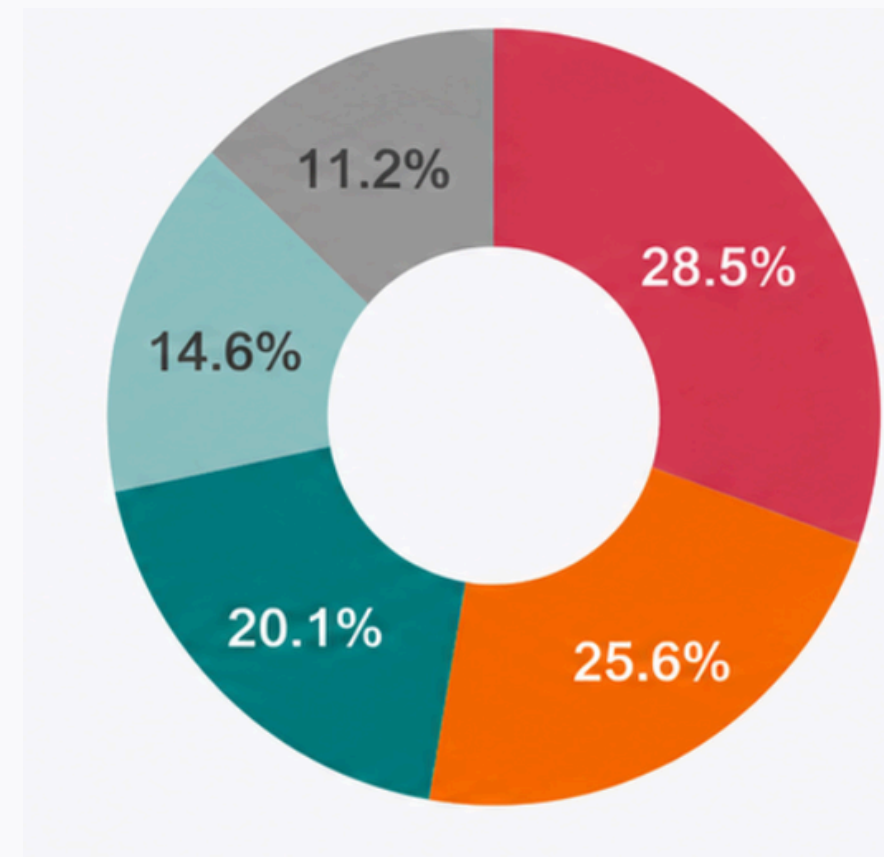
Corporate Power



Innovation Opportunity

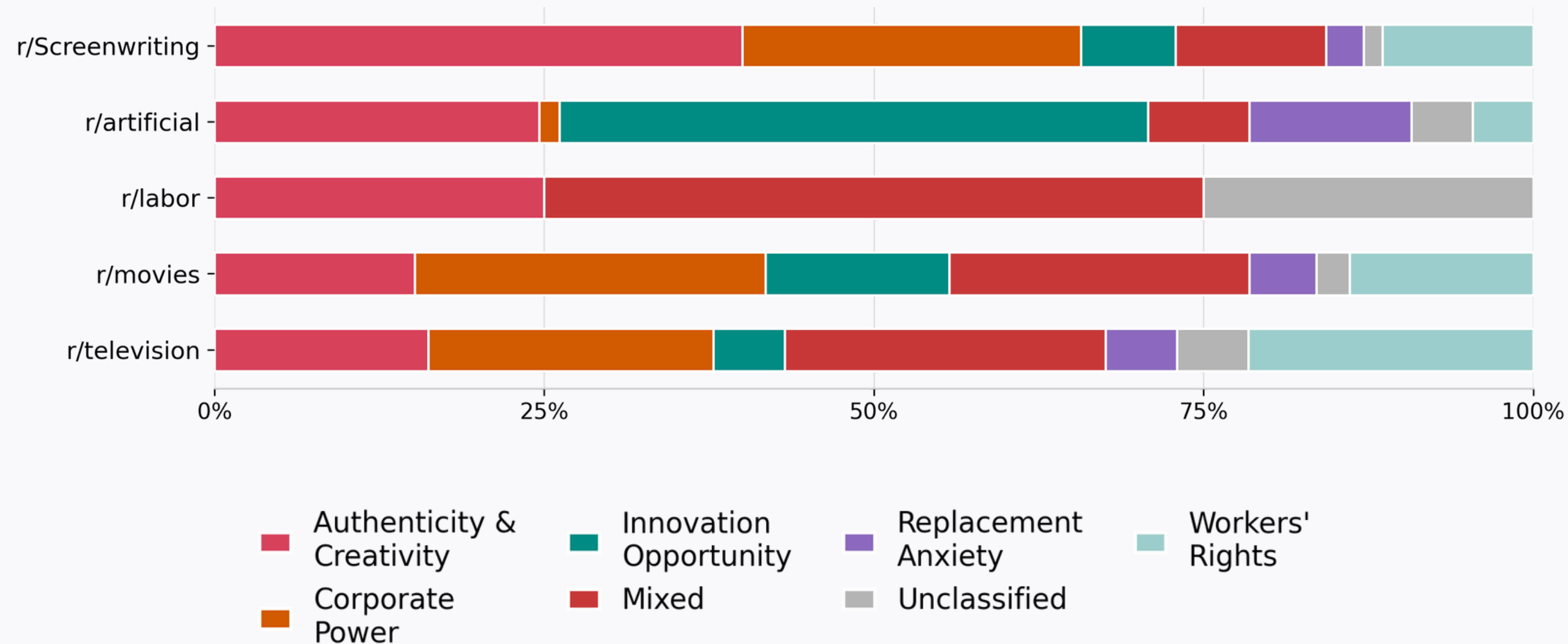
**Dictionary-based
frame classification**

DOMINANT FRAMES IN THE CORPUS



Discussions focus mainly on **creativity, authorship** and **corporate influence**, rather than on job replacement.

Different Subreddits, Different Frames



r/Screenwriting

→ AI as a threat to authorship and creative identity

r/artificial

→ AI as an opportunity for technological progress

r/movies & r/television

→ Multiple perspectives coexist

r/labor

→ AI discussed through broader labor and social concerns

Network Overview



2,684
nodes (users)



3,139
directed edges (replies)

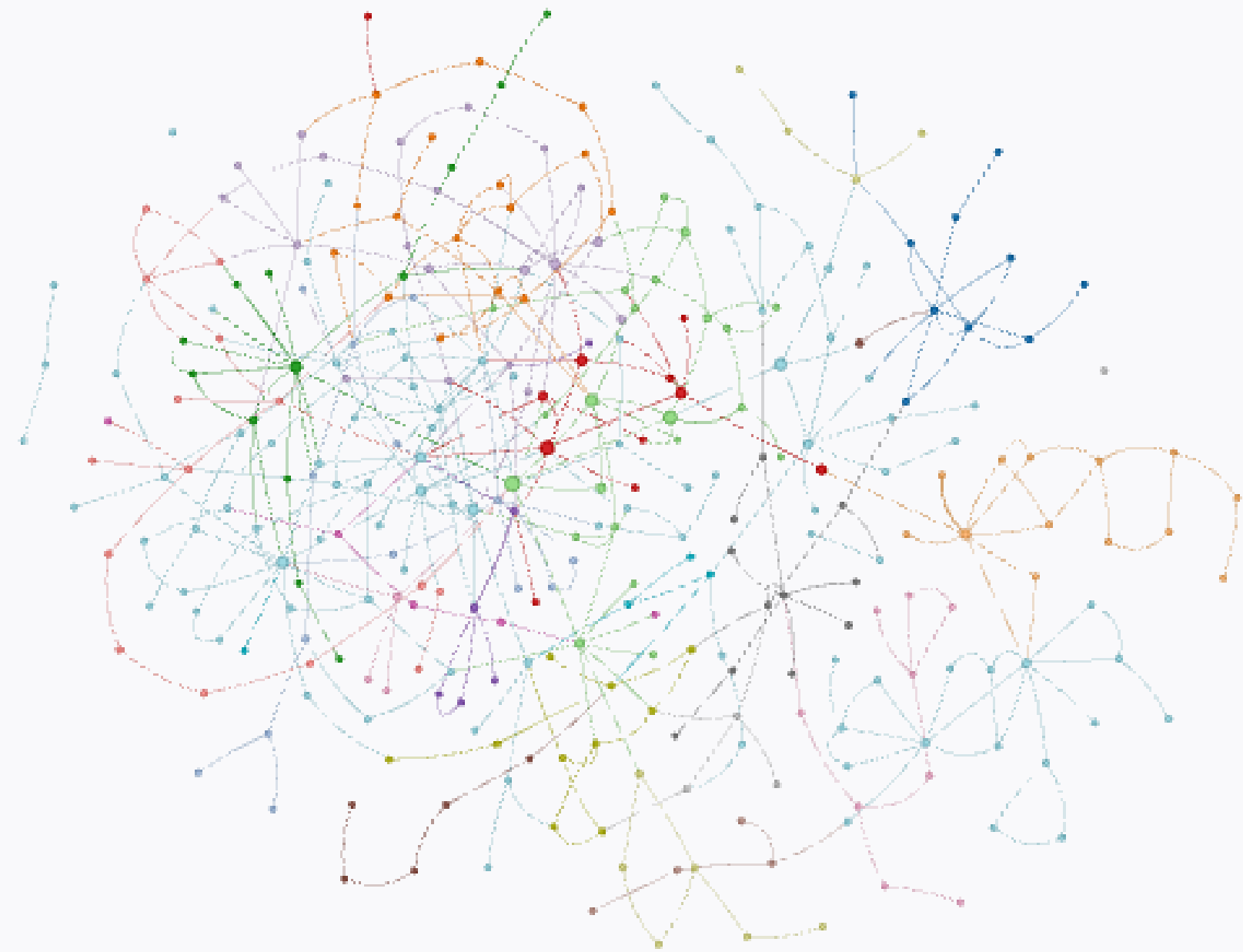


0,000436
density



1,459 users
largest weakly
connected component

Reply interaction network



Users interact through a **sparse reply network** composed of many disconnected conversational clusters

Community Structure



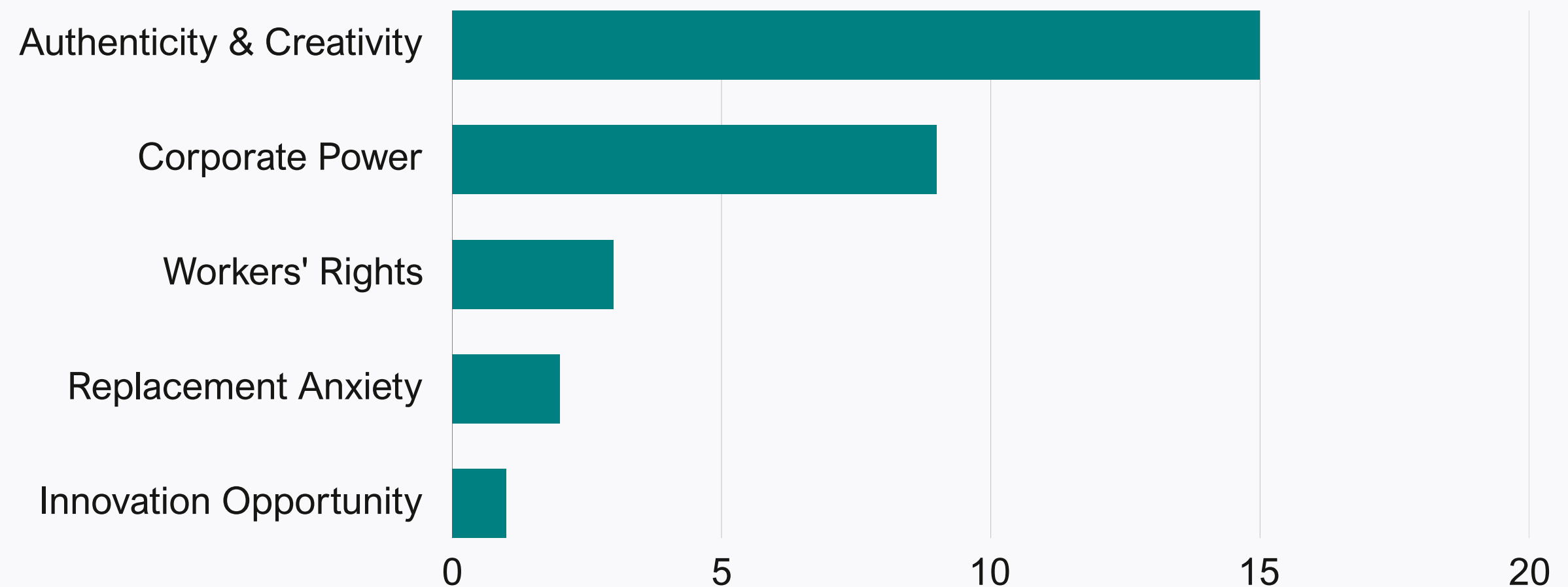
33
communities



0,867
modularity

Interactions occur **mainly within** communities rather than across them

Dominant frame of detected communities



Communities discuss AI through **different interpretative lenses**

Brokers Analysis

Top 10 users by betweenness centrality



spikej

highest betweenness centrality

0.026



9/10
top brokers are active in a single subreddit



8/10
top brokers belong primarily to **r/Screenwriting**

➔ High betweenness reflect **local prominence** within a community, rather than cross-community mediation

Rank	User	Betweenness Centrality	Top Subreddit	N. active subreddits
1	spikej	0.0258	Screenwriting	1
2	sour_skittle	0.0217	Screenwriting	1
3	WilsonEnthusiast	0.0204	Screenwriting	2
4	joet889	0.0169	Screenwriting	1
5	Prince_Jellyfish	0.0164	Screenwriting	1
6	Theaveragegamer12	0.0161	movies	1
7	Penenko	0.0159	Screenwriting	1
8	Craig-D-Griffiths	0.0150	Screenwriting	1
9	Block-Busted	0.0148	artificial	2
10	Extension-Bar5923	0.0129	Screenwriting	1

Temporal Evolution



Communities increase

8 → 27 → 23
pre strike post-strike



Modularity rises

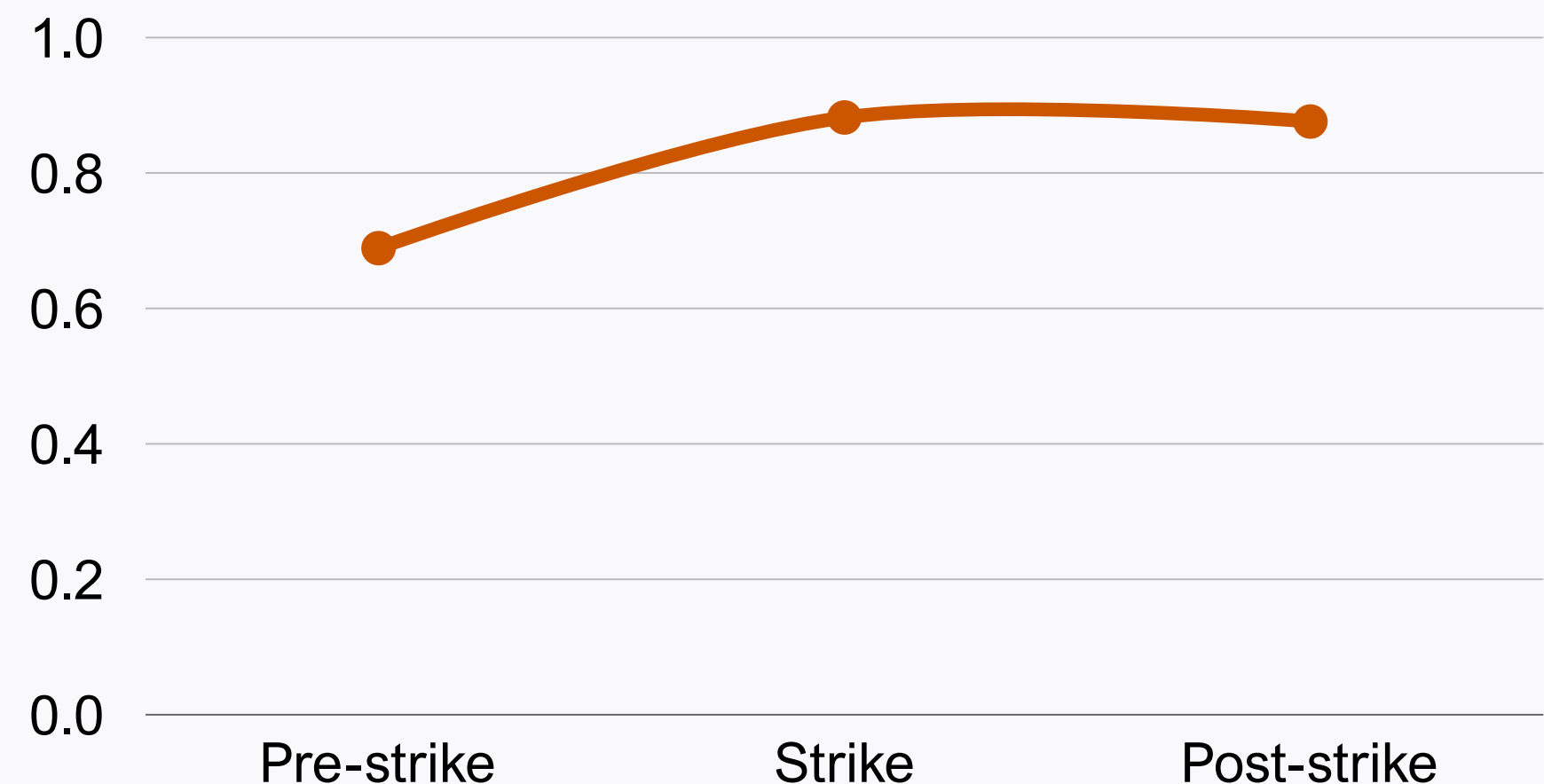
0.689 → 0.882 → 0.876
pre-strike strike post-strike



Cross-subreddit interaction declines

15.5% → 14.6% → 12.4%
pre-strike strike post-strike

Community Segregation Over Time



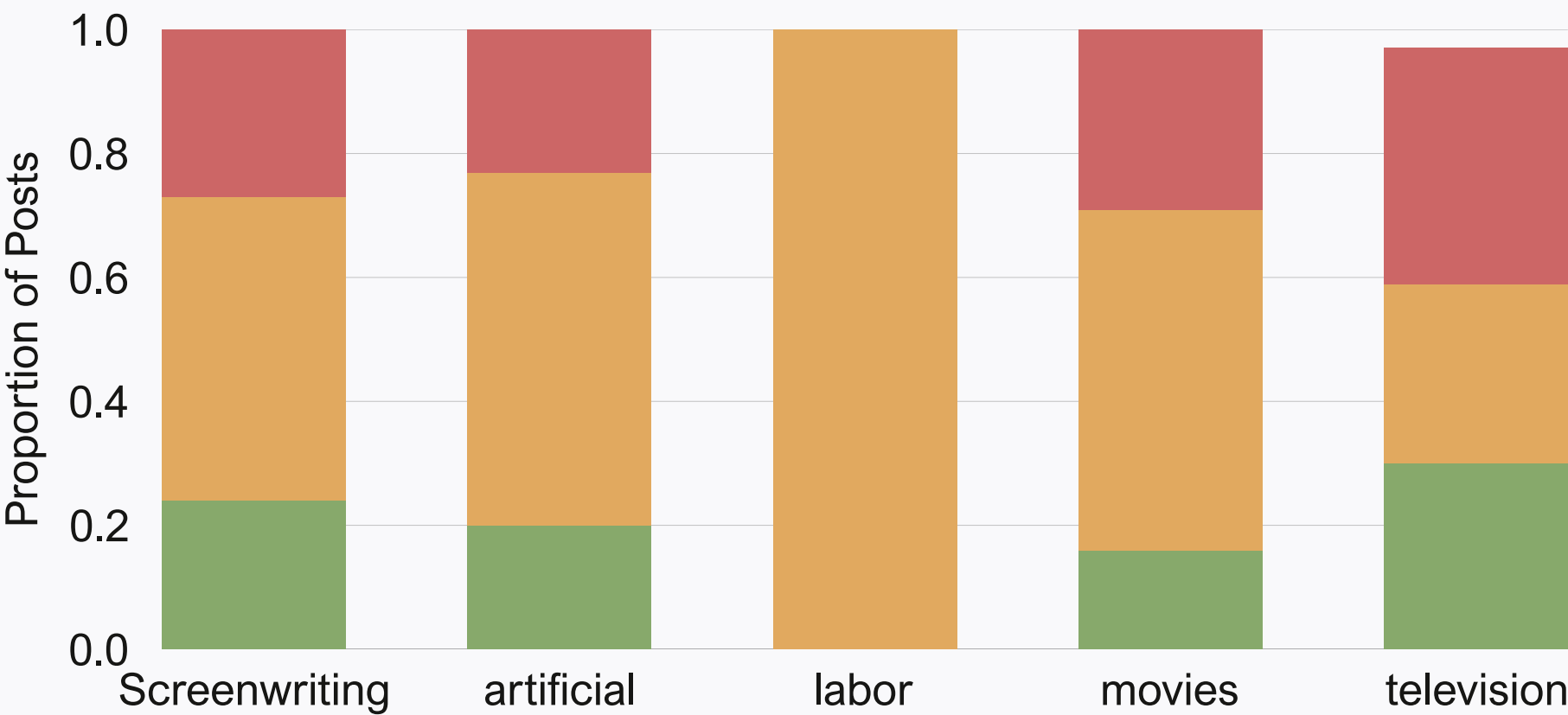
The strike **reinforced community boundaries** rather than creating a more unified discussion

Sentiment: overview and subreddit

How communities felt about AI?

255 posts

RoBERTa-based transformer model:
cardiffnlp/twitter-roberta-base-sentiment



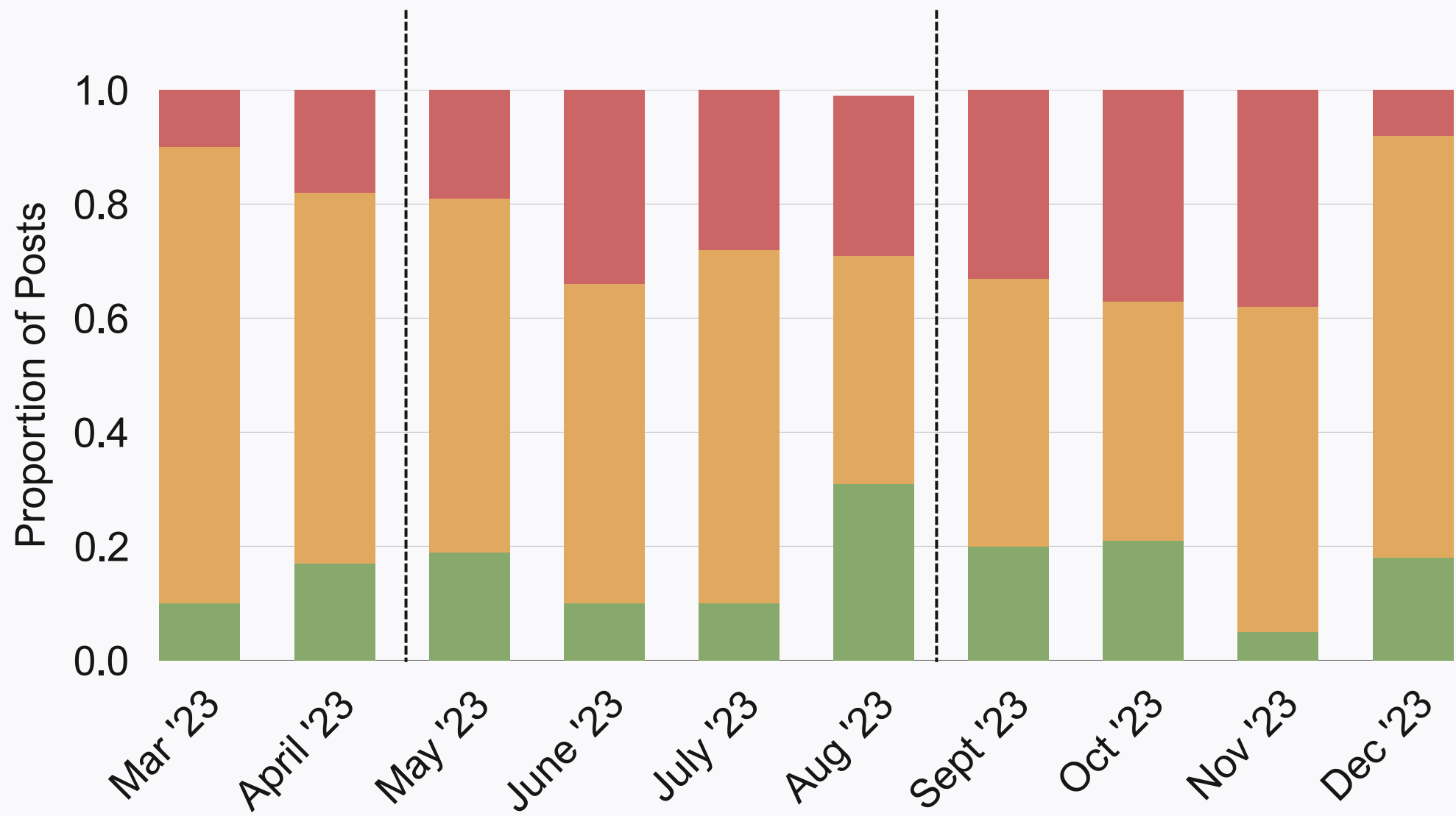
Sentiment Analysis

01	Positive Posts (44)	17.3%
01	Neutral Posts (140)	54.9%
01	Negative Posts (71)	27.8%

Neutral sentiment dominates

Sentiment: temporal evolution

Negative sentiment increases steadily across phases, peaking post-strike rather than during the conflict itself. Concerns about AI remained salient even after the dispute formally ended.



Sentiment distribution

01	Pre-Strike	~14%	~71%	~15%
02	Strike	~18%	~54%	~28%
03	Post-Strike	~16%	~52%	~32%

Different frames centered on different institutional actors, providing additional support for the validity of the frame classification.

01

Workers' Rights → WGA, SAG-AFTRA

02

Innovation Opportunity → OpenAI,
Microsoft, DeepMind

03

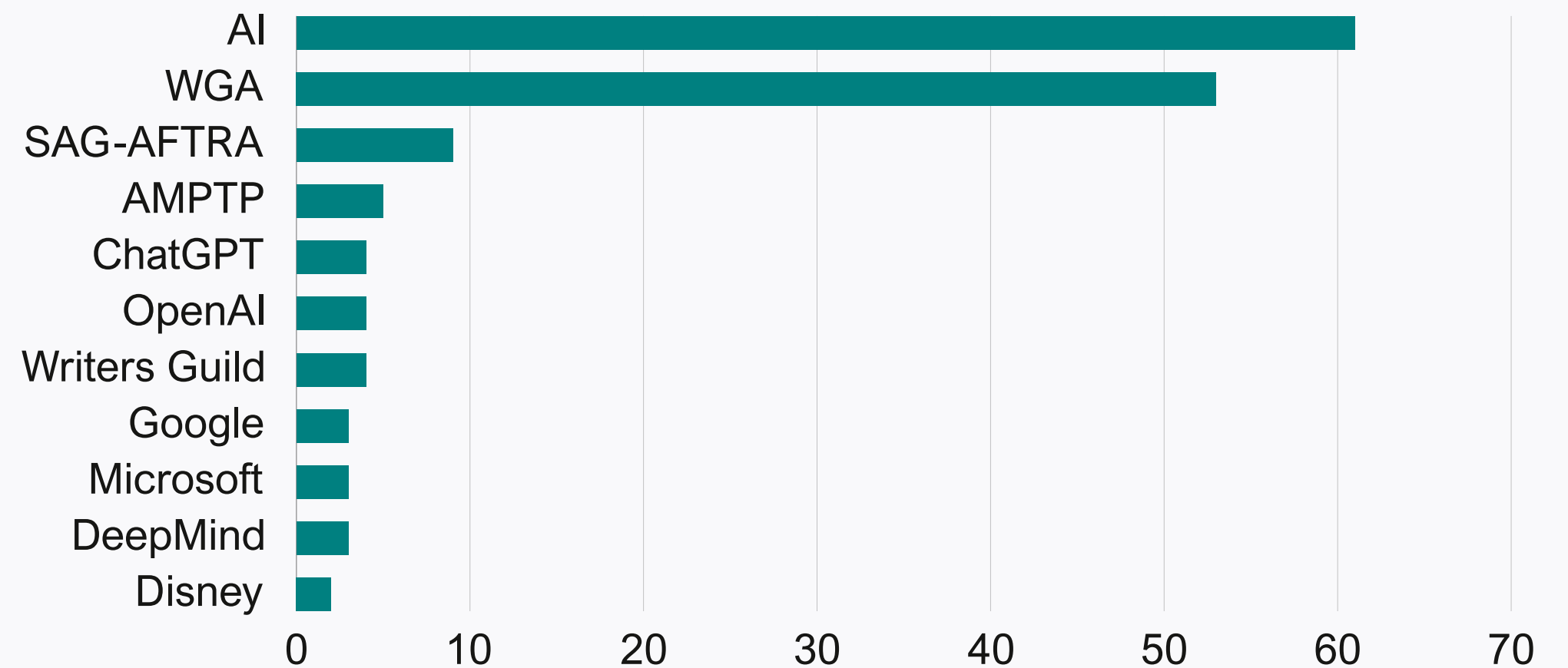
Authenticity & Creativity → mix of
creative and institutional actors

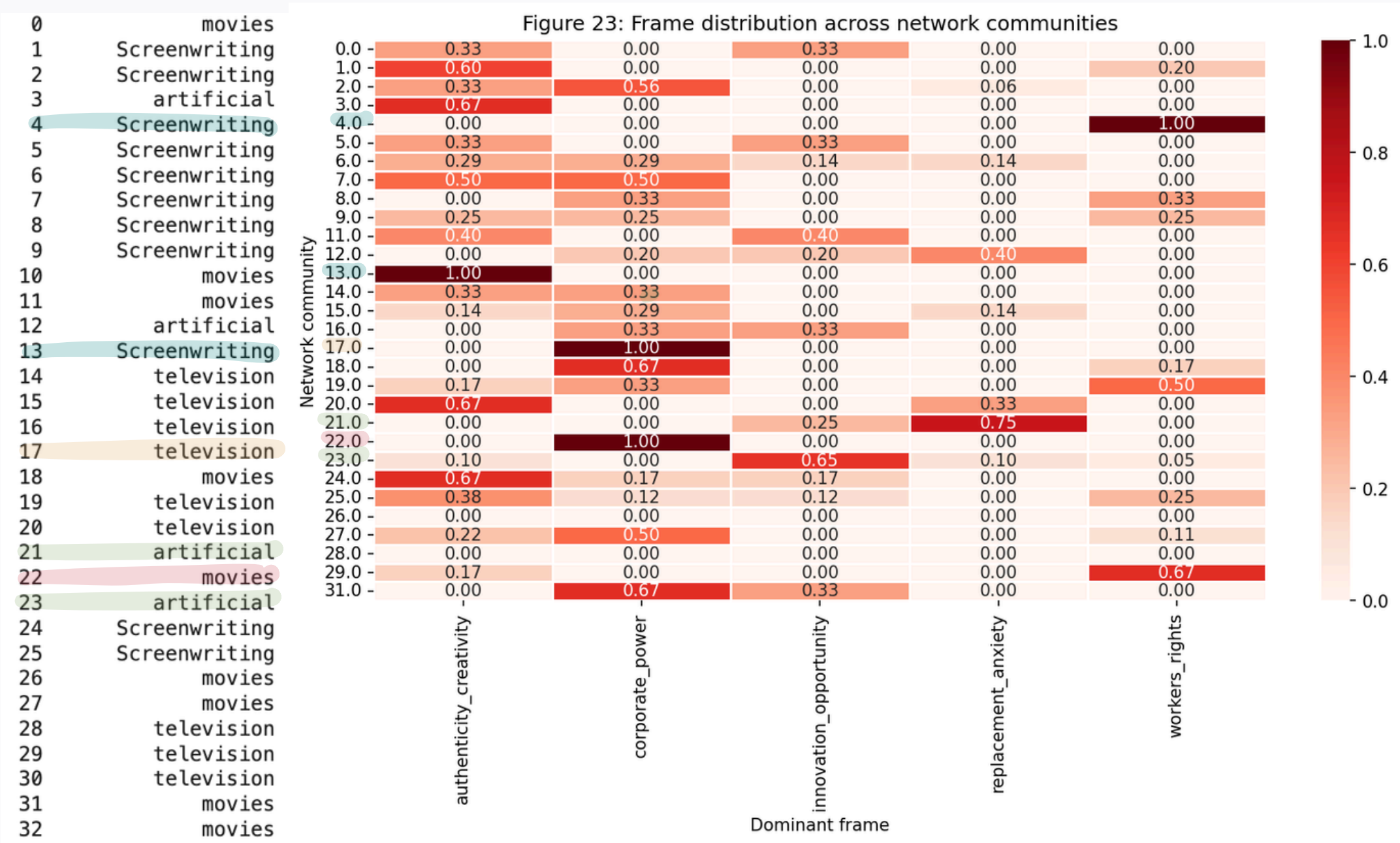
04

Corporate Power → mix of union,
corporate and AI references

Named Entity Recognition

NER applied to 204 posts · spaCy en_core_web_sm





Structure and content aligned

What each interaction cluster discussed?

117 posts mapped to network communities
Communities that interacted less frequently also tended to emphasize different dominant frames, network boundaries and interpretive boundaries reinforced one another

r/Screenwriting → Worker’s Rights, Authenticity & Creativity
r/artificial → Replacement anxiety, Innovation Opportunity
r/television & r/movies→ Corporate Power

So, how did online communities frame AI during the Hollywood Strike?

Community identity shaped AI framing. Different subreddits emphasized creativity, labor, corporate power, or innovation. These differences were consistent across frames, sentiment, and entity references.

The discussion remained fragmented. Users interacted mostly within their own communities rather than across them. Cross-subreddit interaction declined from 15.5% pre-strike to 12.4% post-strike, the conflict reinforced rather than dissolved community boundaries

AI concerns persisted after the strike. Negative sentiment remained high even after the dispute ended, suggesting that the formal resolution redirected rather than resolved underlying anxieties.



The Hollywood Strike was not debated but it was experienced differently by different communities, in parallel.



Thank You.